Cloud

Module 5 - Container

Input of this Module

- https://learning.oreilly.com/playlists/43827098-7a87-435e-823e-736586b5694c
  05-container
- https://access.redhat.com/documentation/de-de/red_hat_enterprise_linux/8/html/managing_monitoring_and_updating_the_kernel/using-cgroups-v2-to-control-distribution-of-cpu-time-for-applications_managing-monitoring-and-updating-the-kernel
- https://wizardzines.com/comics/
- Several Blogposts

slido

Please download and install the Slido app on all computers you use

What is a container?

ⓘ Start presenting to display the poll results on this slide.

Containers are disruptive…

Performance:
- Minimal overhead of process

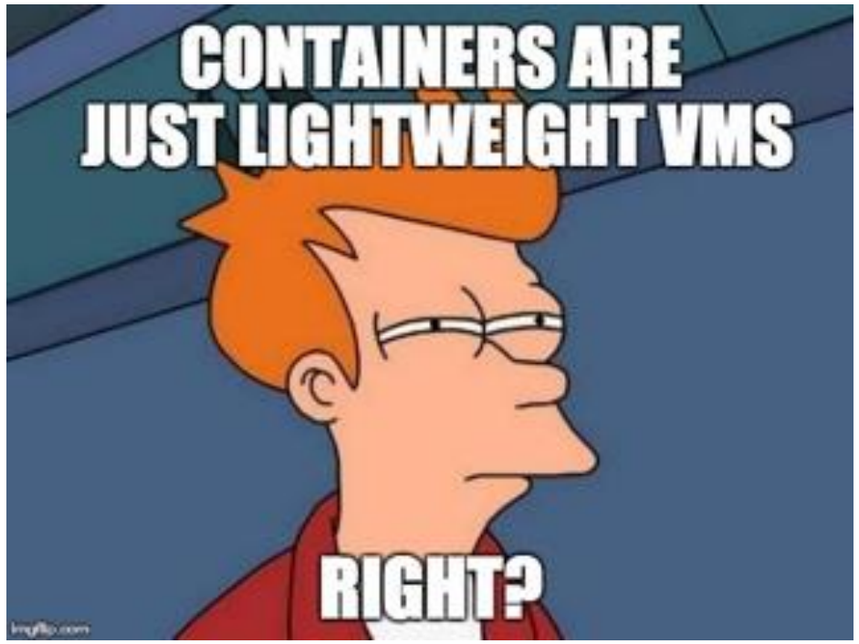
Easy scalability:
- Easy to scale horizontally without provisioning, startup, etc

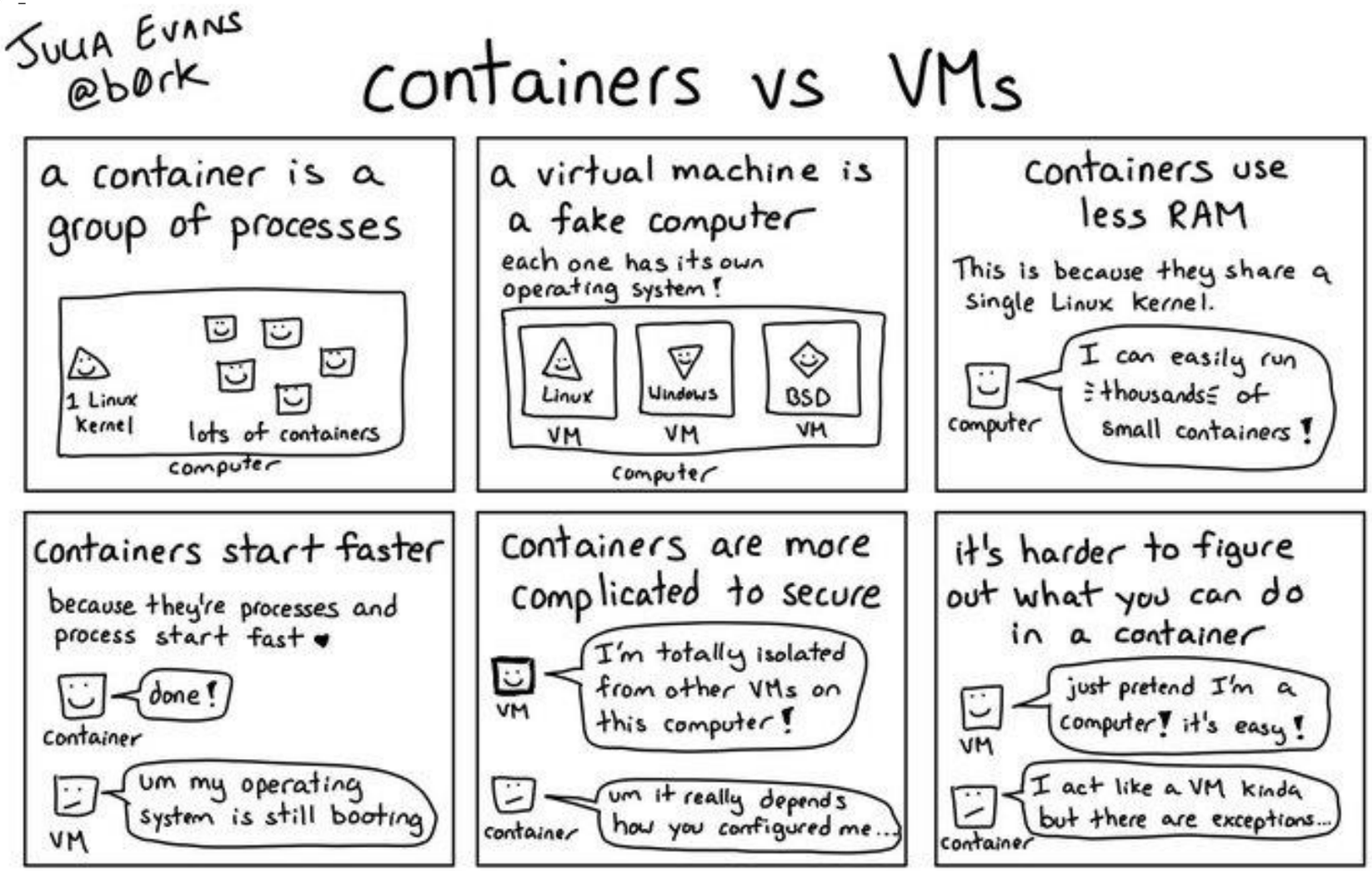
Portability:
- Easy transfer of container to different hosts via http/s

Consistency:
- Self contained environment to run applications

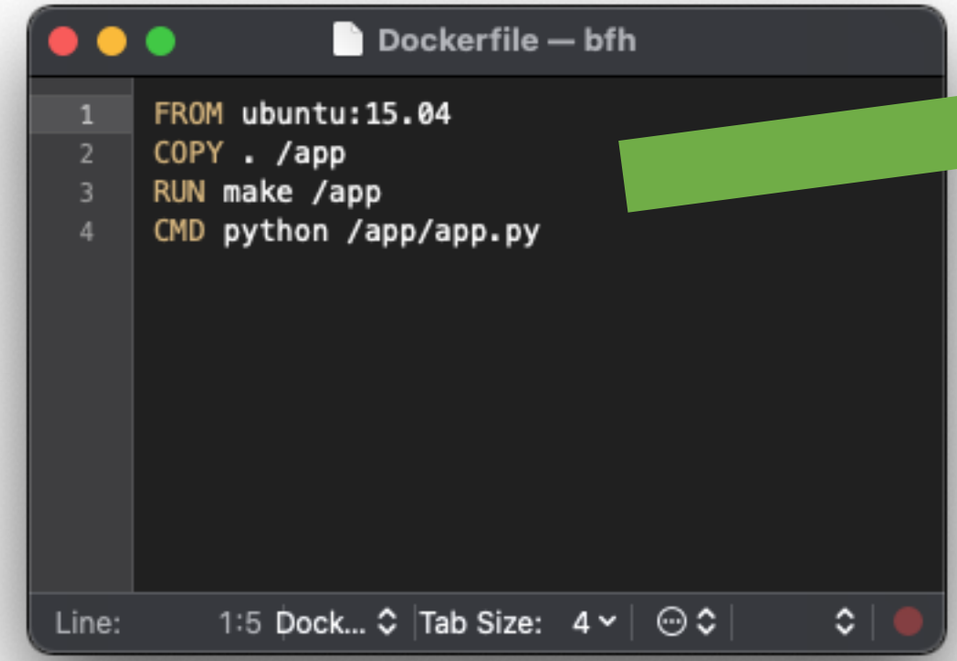
Free and wide Ecosystem:
- Tools, platform is free and easy to use

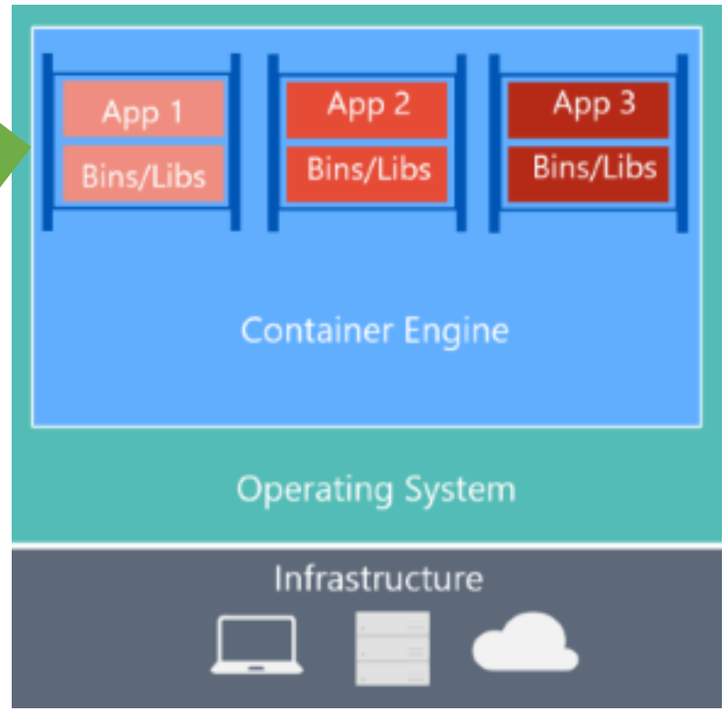
…but what are containers?





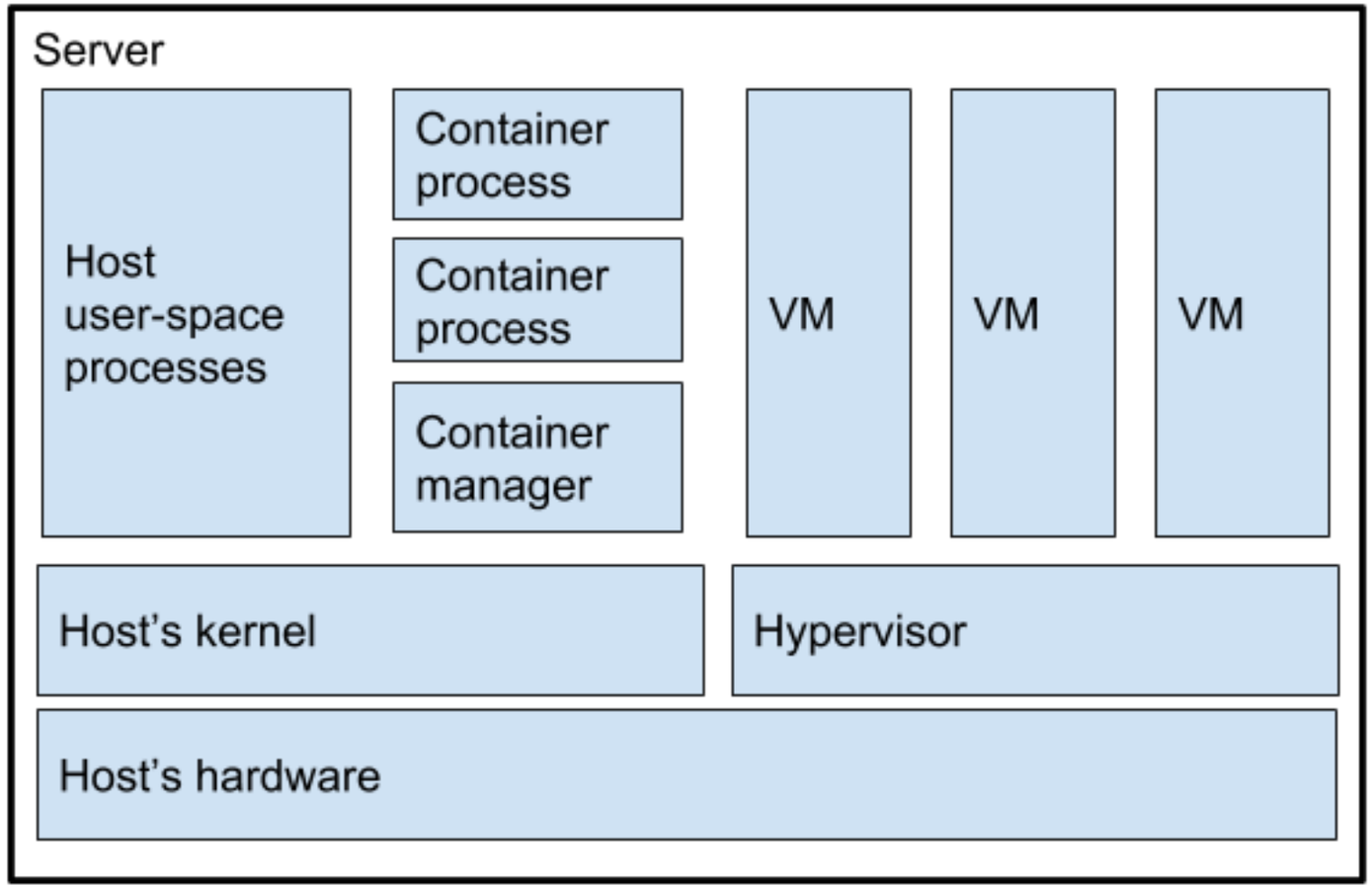
We all use Docker…





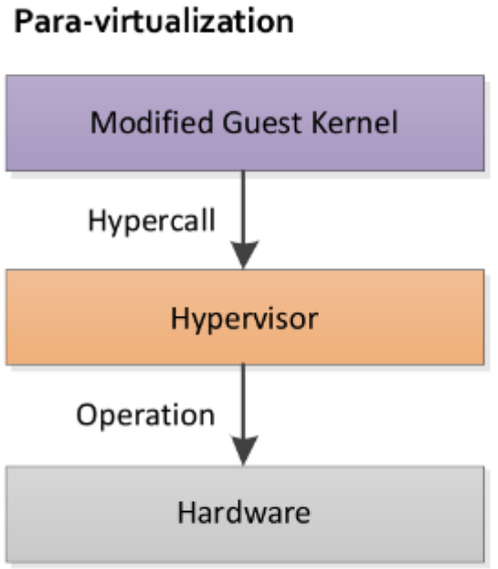
https://docs.docker.com/storage/storagedriver/images/container-layers.jpg

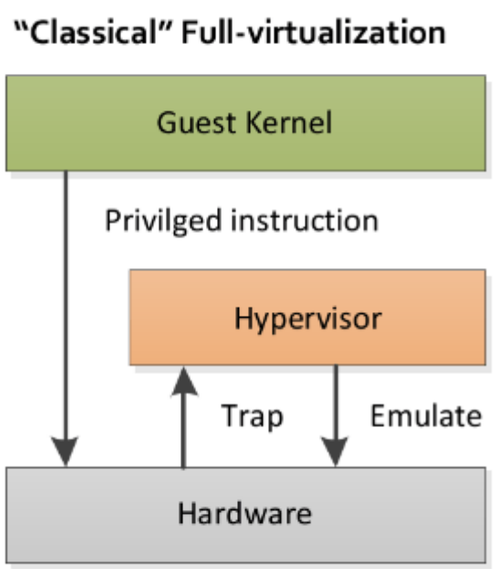
Containerization VS Virtual Machines



Recap: Paravirtualized Environments

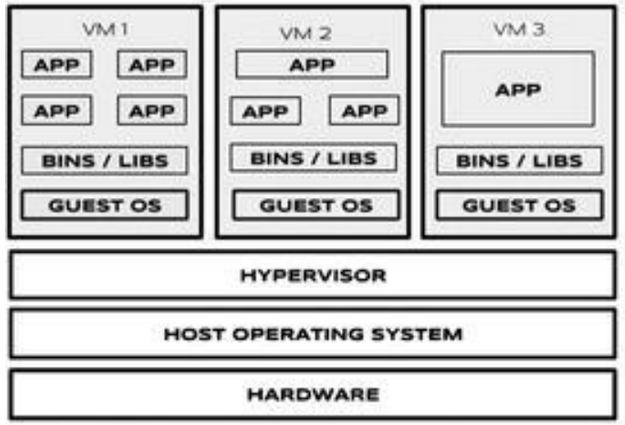
- Guests entirely rely on hypervisor-enabled Kernel
  - In Practice: they share the same kernel
- Calls can be directly made to hypervisor (“hypercall”)
  - no interception / emulation necessary
- +: Less overhead, very efficient
- -: relys on kernel of host
- Famous products: lxc, xen, solaris zones



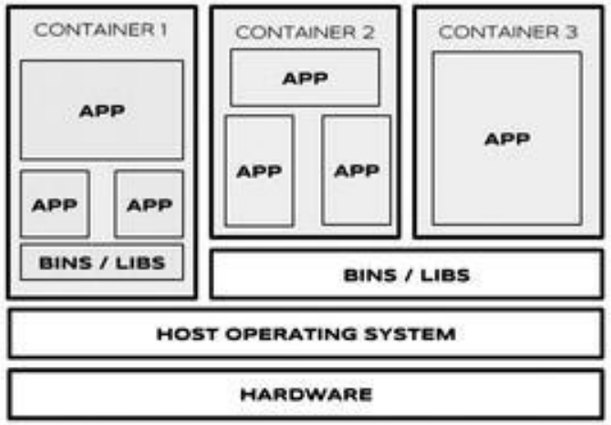


Special Kernel needed…but Is this really necessary for containers?

Containerization



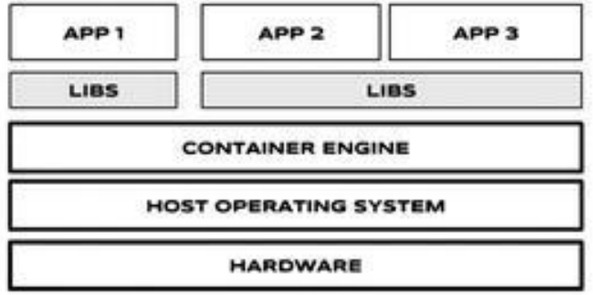
Virtual Machines on Host



Linux Containers (lxc) or Operating System Level Containers

Different Levels:

- Application Level:
  Isolated Space where minimal processes run
- Operating System Level:
  Entire operating systems runs in isolated space



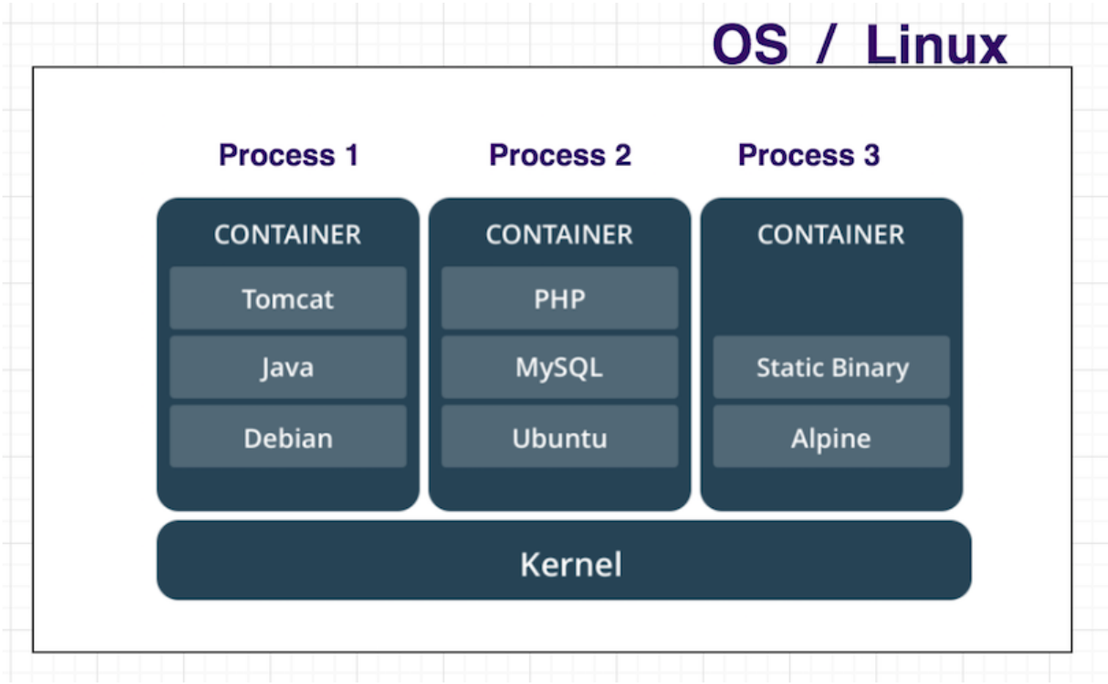
Application Level Containers

Under the hood: What is a container?

A container is a set of fixed defined processes.

Logical consequences are:

- Container have a lifecycle similar to a process.
- Containers have the same overhead than processes
- Containers have no additional start time (e.g. boot time)
- Containers leverage directly from kernel features for process handling / isolation
- Nevertheless, Containers are portable



Such containers are ooooooooooooold

Chroot:

- Still widely in use

LXC:

- relies on Kernel Modules starting 2.6.24
- Cgroups / chroot / namespaces

Docker:

- Started as a convenient interface for lxc

Year	Technology	First Introduced in OS
1982	Chroot	Unix-like operating systems
2000	Jail	FreeBSD
2000	Virtuozzo containers	Linux, Windows (Parallels Inc. version)
2001	Linux VServer	Linux, Windows
2004	Solaris containers (zones)	Sun Solaris, Open Solaris
2005	OpenVZ	Linux (open source version of Virtuozzo)
2008	LXC	Linux
2013	Docker	Linux, FreeBSD, Windows

How to implement a Container from scratch?

Filesystem should be isolated in the container:

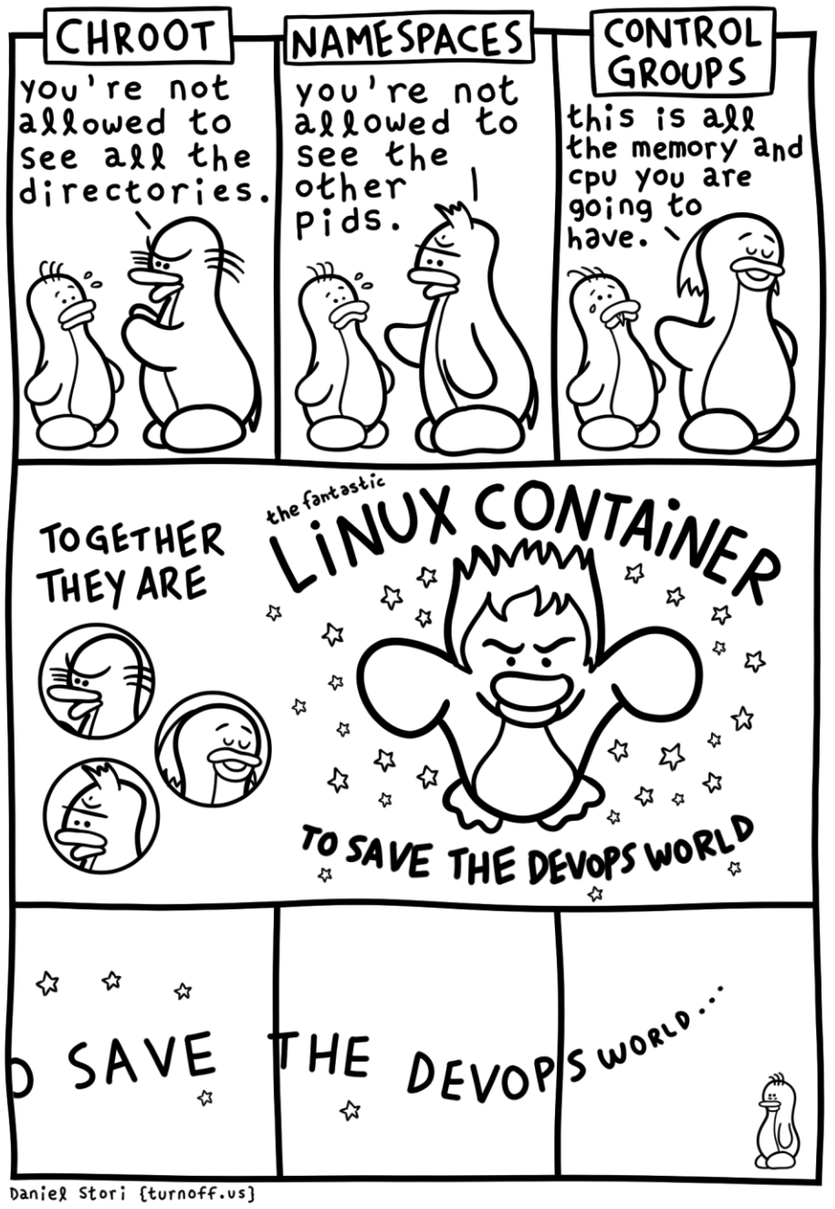
→ Chroot

Processes should look like they are only visible in the container:

→ Namespaces

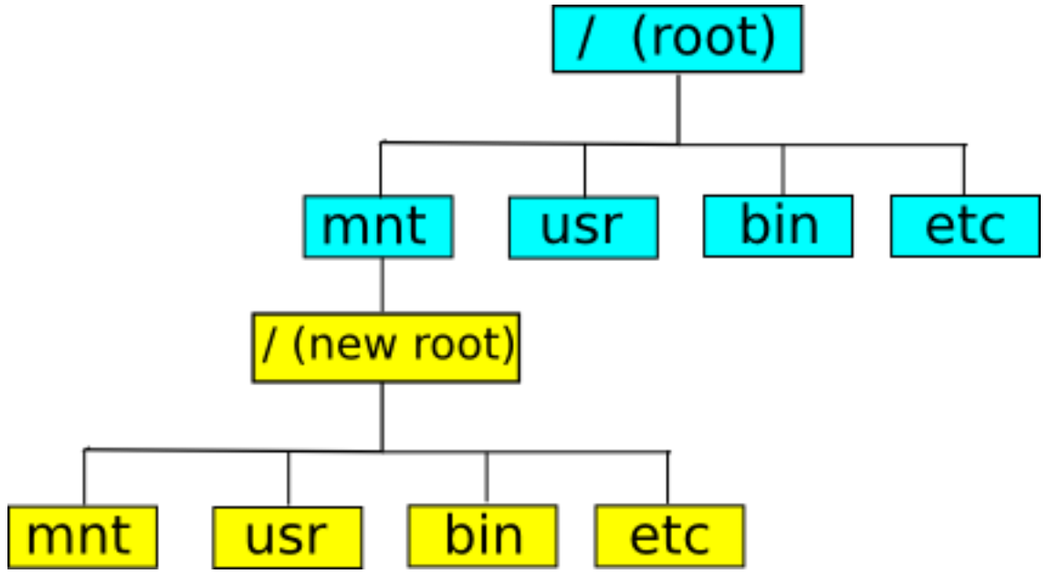
Resources should be restricted to the container:

→ cgroups



Chroot

- Chroot = change root
- Sets actually the root directory including all paths according to the File Hierarchy Standard to a non-root folder (!= /)
- Combined with Mount Namespaces, process can act entirely isolated.



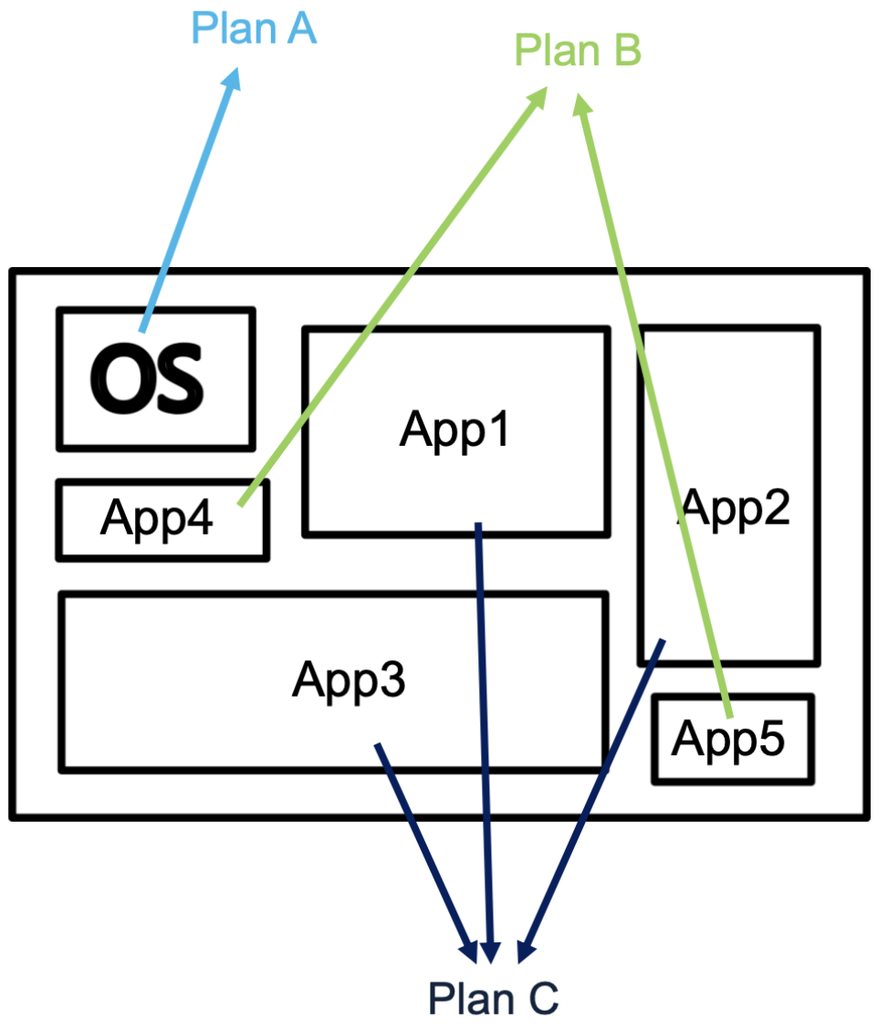
Cgroups

Different plans demand different resources.

App specs are according to cgroups:

- CPU time
- System memory
- Disk bandwidth
- Network bandwidth
- Monitoring

Cgroups (Control Groups) are organized in an hierarchy and summarize processes

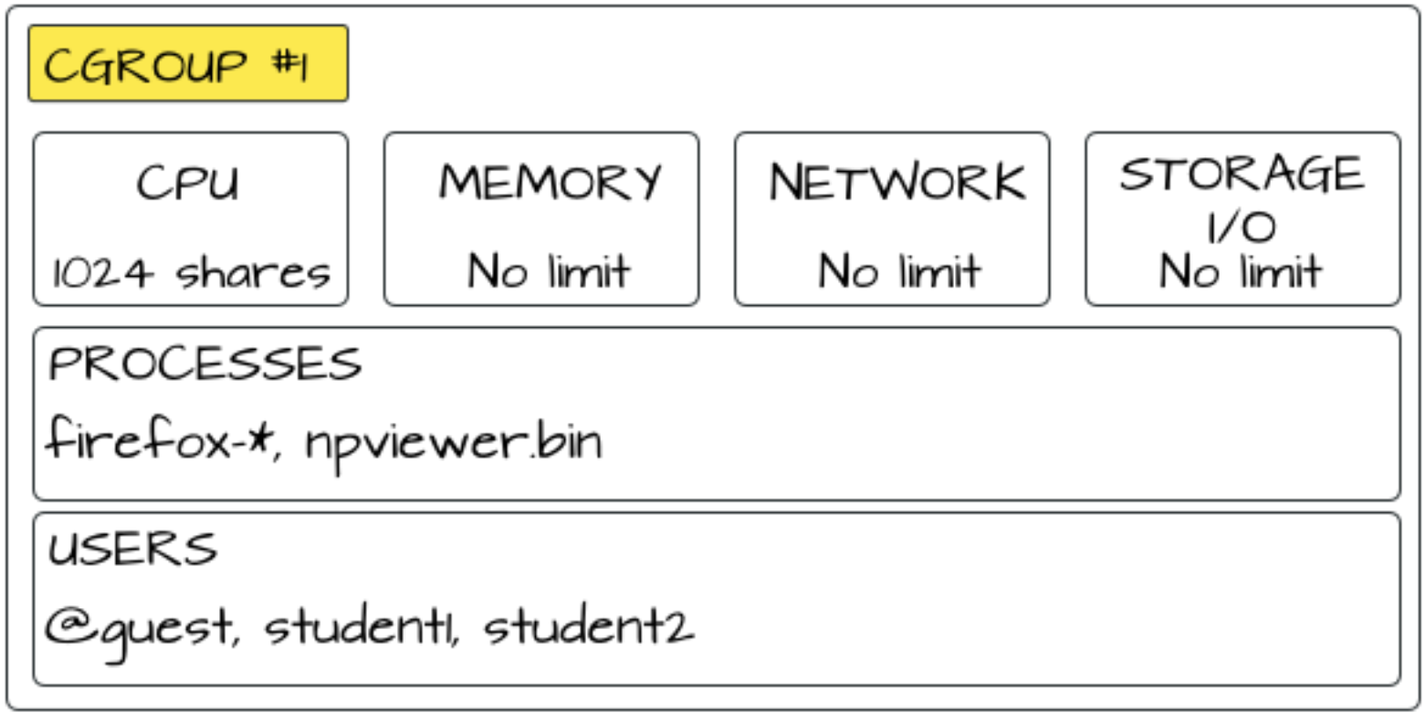


What does a cgroup look like?

Different levels for different resources.

Refer to https://docs.kernel.org/admin-guide/cgroup-v2.html for all possibilities.

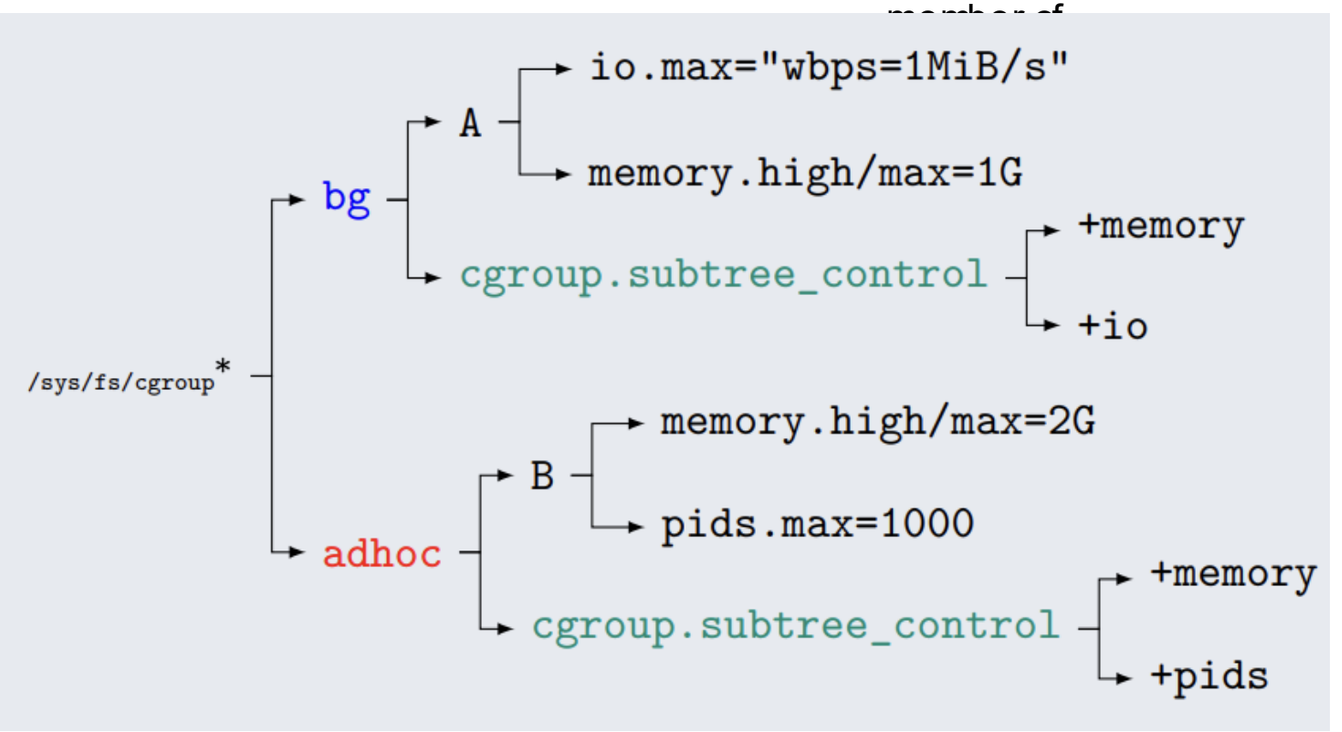
Beware of cgroup.v1 vs cgroup.v2

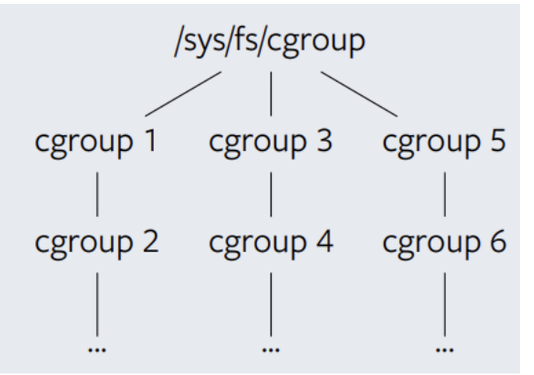


Hierarchy on file system

VFS under `/sys/fs/cgroup`

- Each cgroup contains rules as files
- Each cgroup can be enabled for dedicated rules
- A cgroup can contain other cgroups





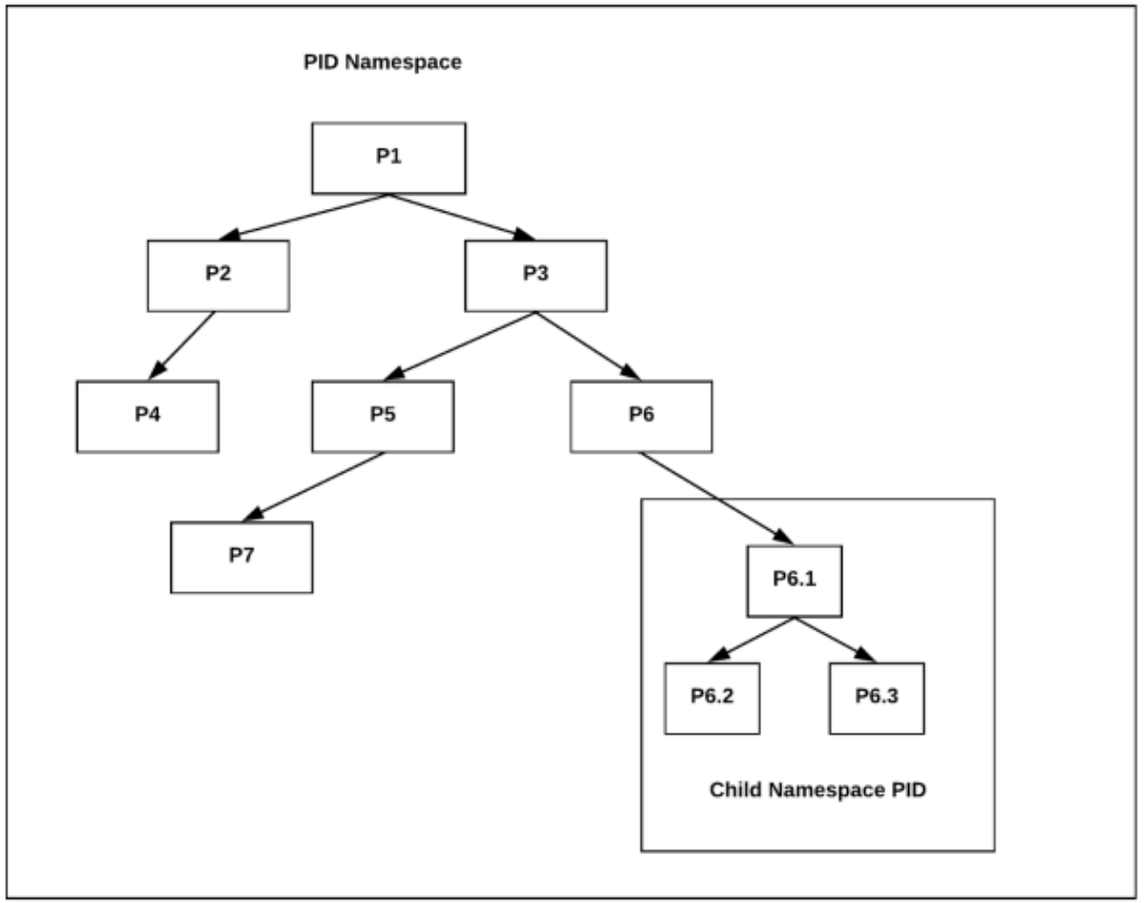
Namespaces

VFS under `/proc/PROCESS`

- Each process contains all necessary information about itself
  - E.g. adaption of cgroup

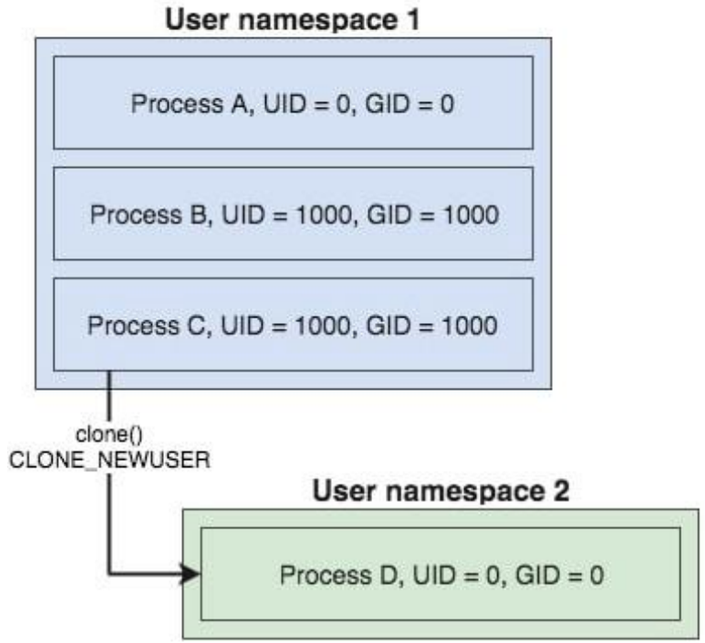
Different system calls to control:

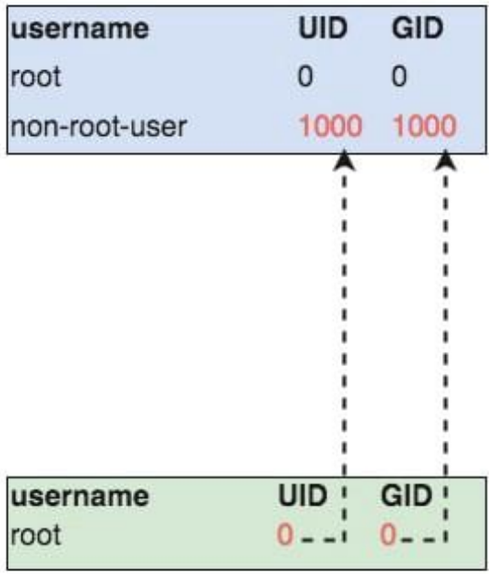
- clone(): This creates a new process and attaches it to a new specified namespace
- unshare(): This attaches the current process to a new specified namespace
- setns(): This attaches a process to an already existing namespace



Namespaces

- Mapping of UIDs/GIDs between Child-Id and global id possible:
  see `/proc/<PID>/uid_map` and `/proc/<PID>/gid_map`
- User in child process assume it is root



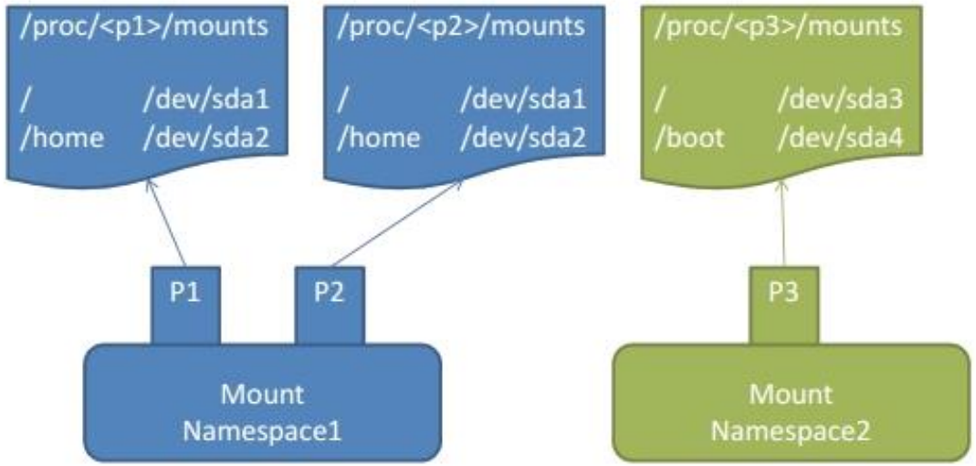


Many different namespaces

- **PID namespace**: isolation of the system process tree;
- **NET namespace**: isolation of the host network stack;
- **MNT namespace**: isolation of host filesystem mount points;
- **UTS namespace**: isolation of hostname;
- **IPC namespace**: isolation for interprocess communication utilities (shared segments, semaphores);
- **USER namespace**: isolation of system users IDs;
- **CGROUP namespace**: isolation of the virtual cgroup filesystem of the host.

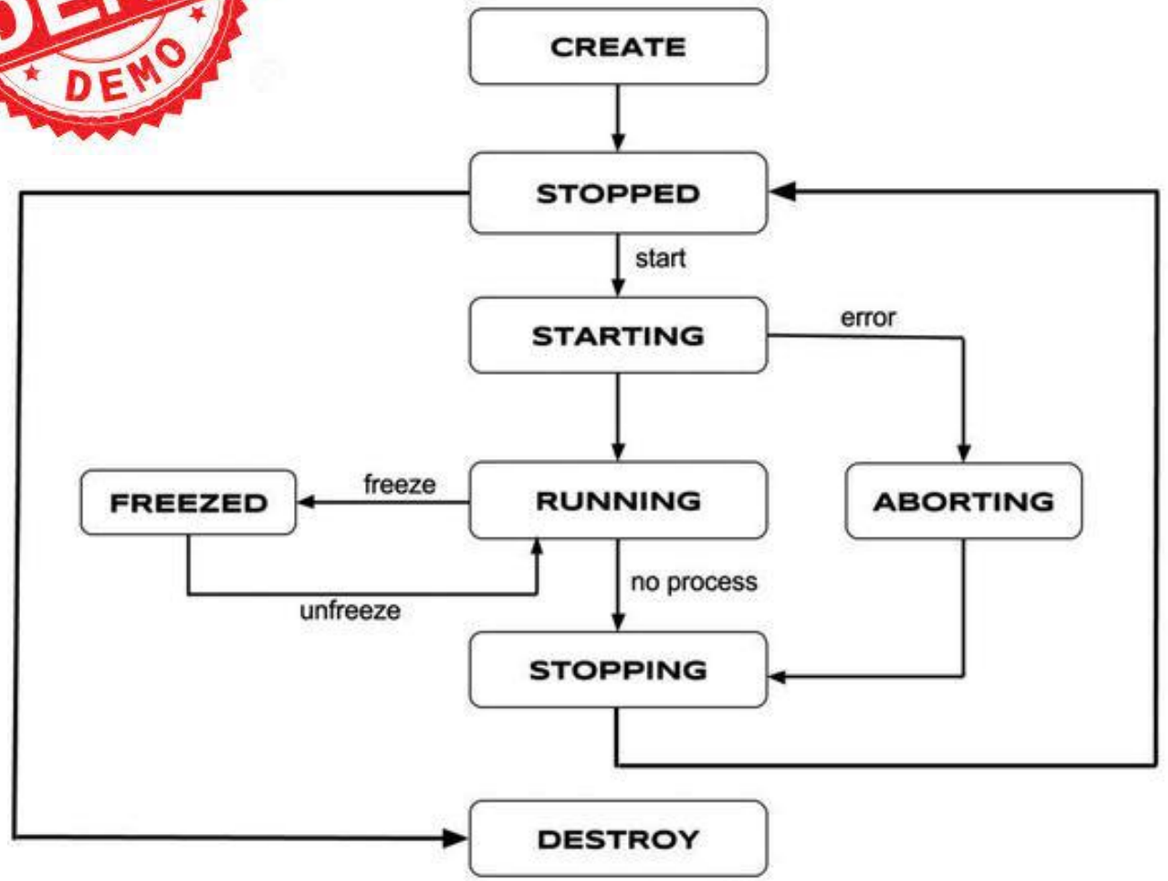
Mount Namespaces

- Namespaces can have dedicated mounts
  - Processes in these namespaces are isolated with respect to the mounted block devices
- Shared mounts between namesspaces are possible though



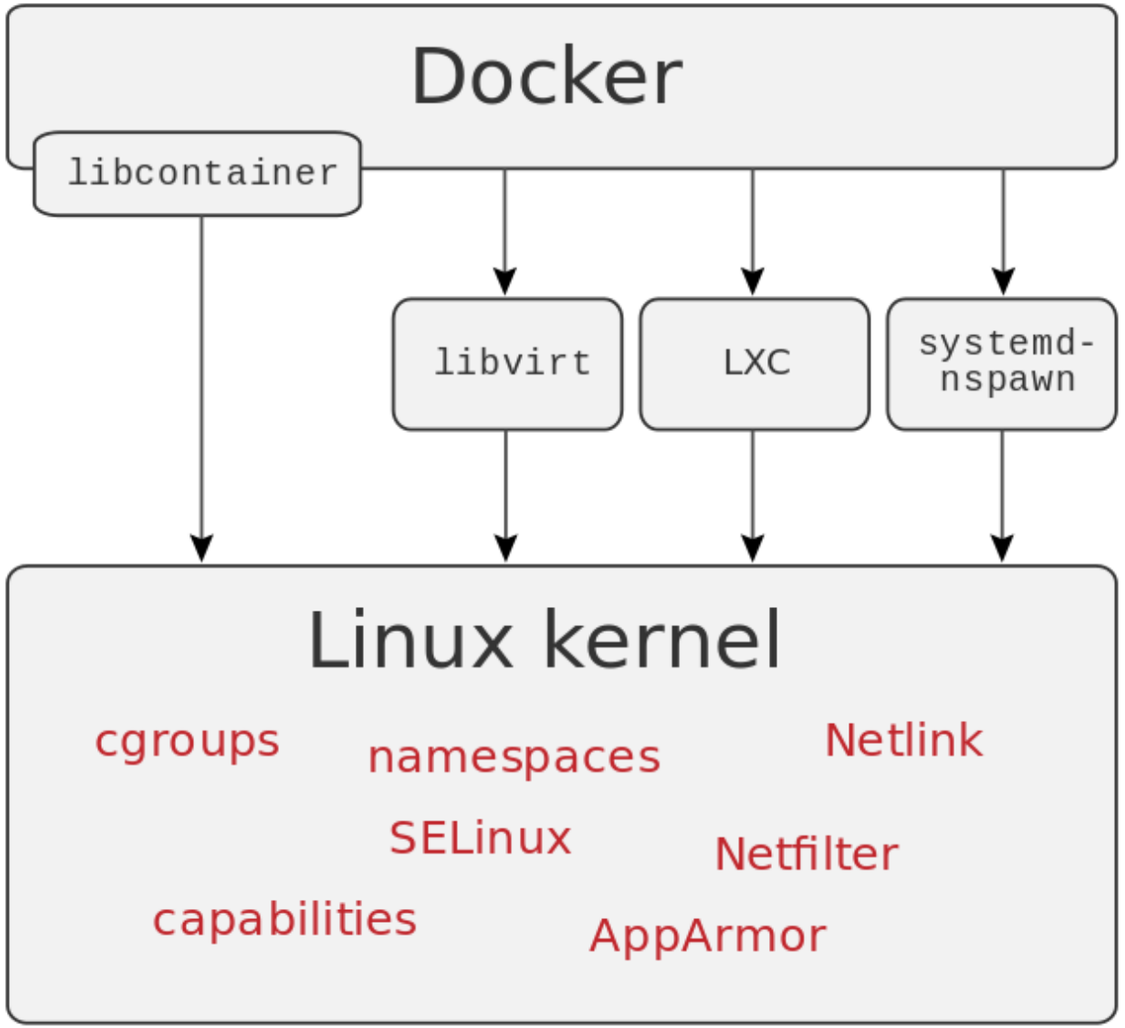
Putting it all together, lxc/lxd

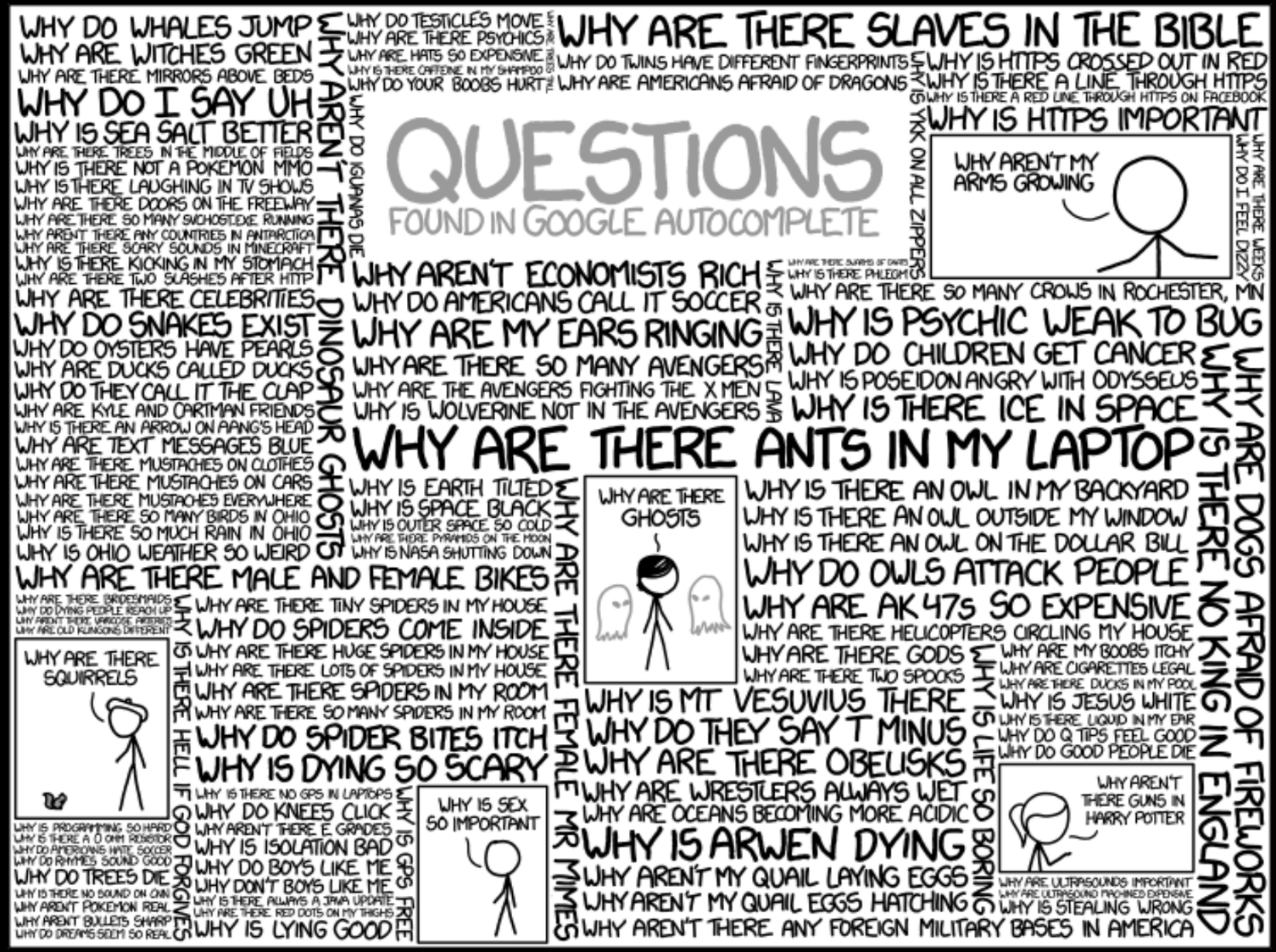
`1.lxc-create`: Create a container with the given OS template and options.
`2.lxc-start`: Start running the container that was just created.
`3.lxc-ls`: List all the containers in the host system.
`4.lxc-attach`: Get a default shell session inside the container.
`5.lxc-stop`: Stop the running container, just like powering off a machine.
`6.lxc-destroy`: If the container will no longer be used, then destroy it.



Docker

Docker (as well as all containers) do not really virtualize / emulate:
They rely on kernel features only provided via different libs




QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SYCHOSTICKS RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REARM UP
WHY AREN'T THERE VARICOSE AIRBIKES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY DO IGUANAS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE LAVA
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS SEX SO IMPORTANT
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY AREN'T THERE DINOSAUR GHOSTS

Summarizing Questions

- What technologies / kernel features are actually needed for having containers?
- Do container emulate? If so, why is it necessary? If not, why is it not necessary?
- I want to restrict the memory usage of a container: what feature of the kernel enables me to do so?
- I want to install software software in my container: why do mount namespaces protect my host?
- Why do PID-namespace the usage of a single root-process in an application container?